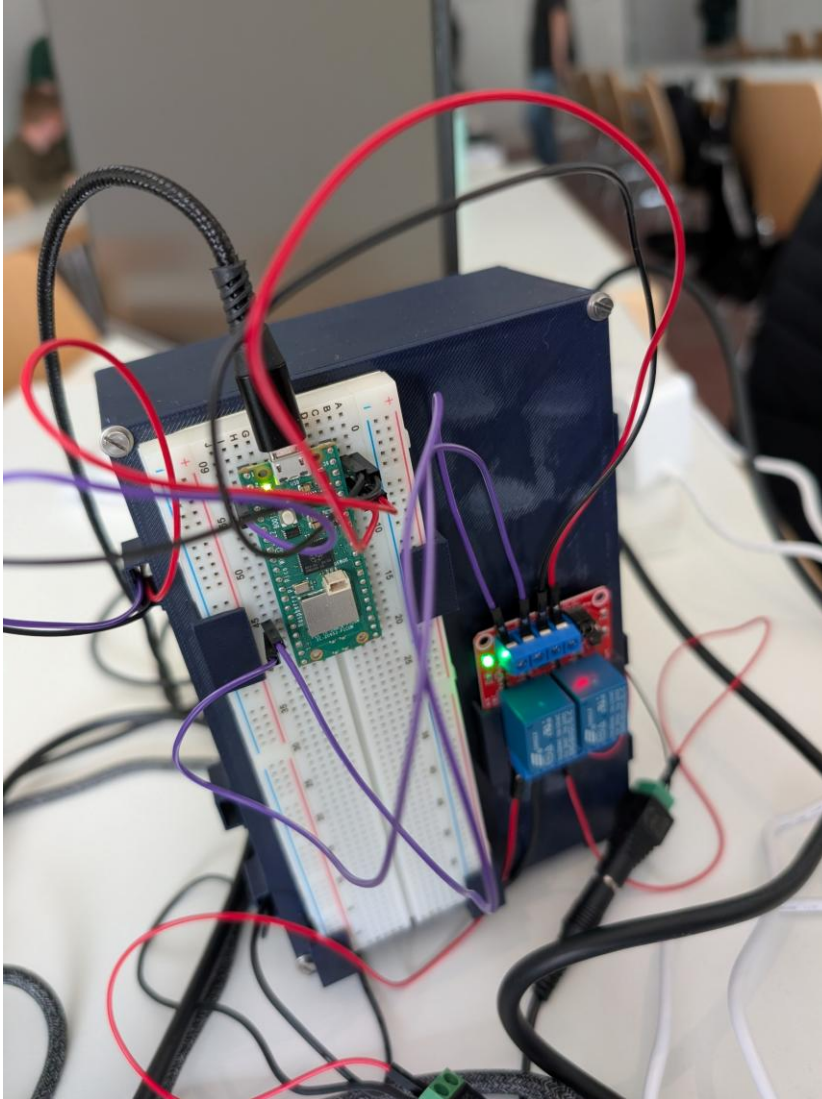


Phase 3

Anwendungsprojekt Informatik



Struktur

1. Meilenstein

2. Hardware

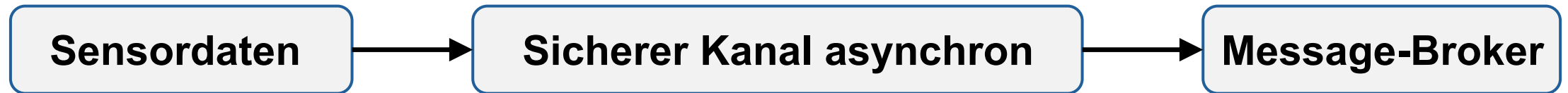
3. Docker

4. CA

5. MQTT

1. Meilenstein

Sichere Übertragung der Sensordaten an MQTT



1. Meilenstein

Benötigte Teilschritte

Backend-Server

- Docker + Docker-Bridge
- Message-Broker als Docker Container

Certificate Authority (CA)

- CA auf Backend-Server
- Root-Zertifikat für CA
- Server-Zertifikat für MQTT-Broker

1. Meilenstein

Benötigte Teilschritte

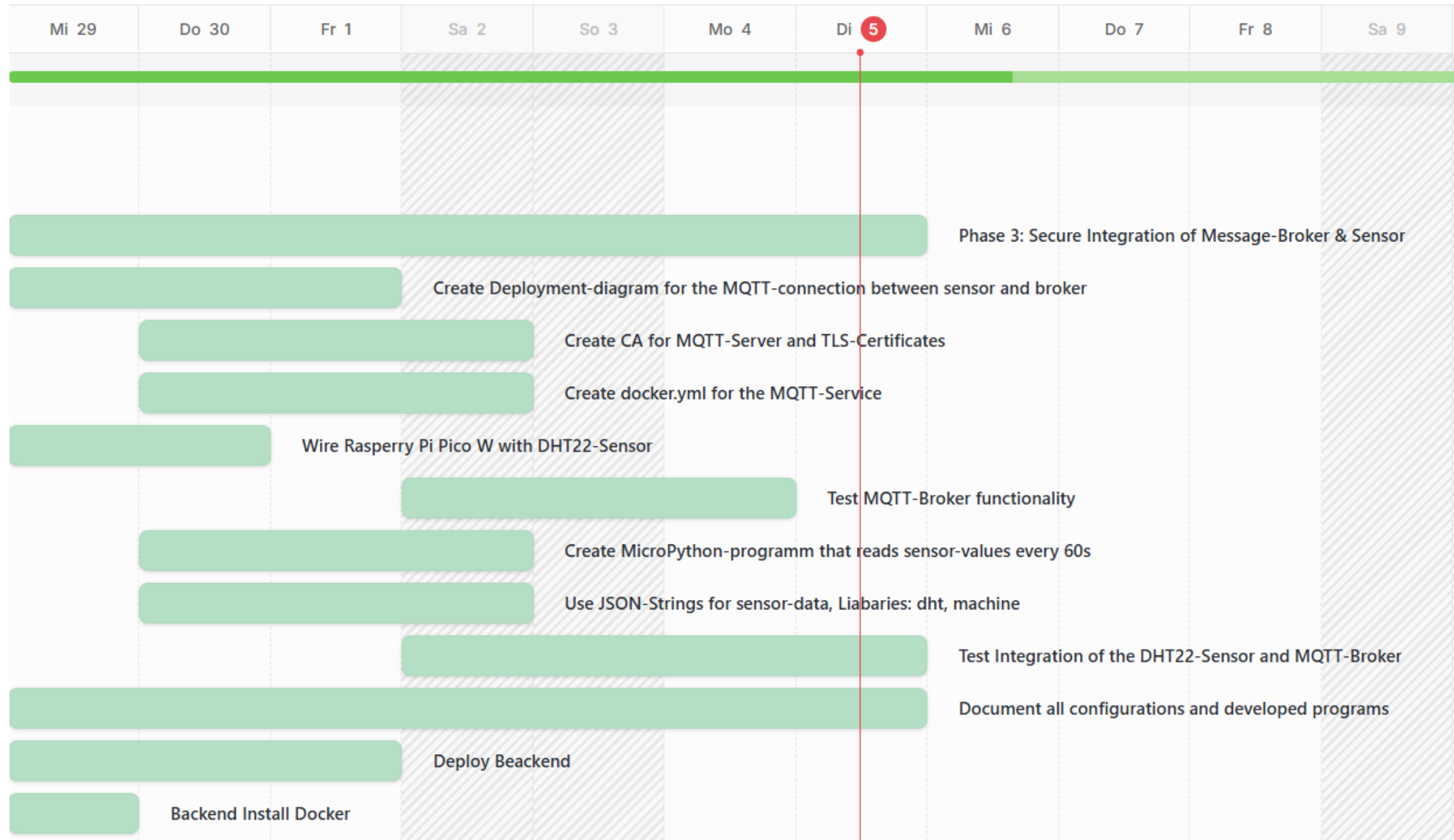
Mikro Python Programm

- Alle 60 Sekunden DHT22 auslesen
- Topic 'sensor/data'
- Daten als JSON-String
- Authentifizierung

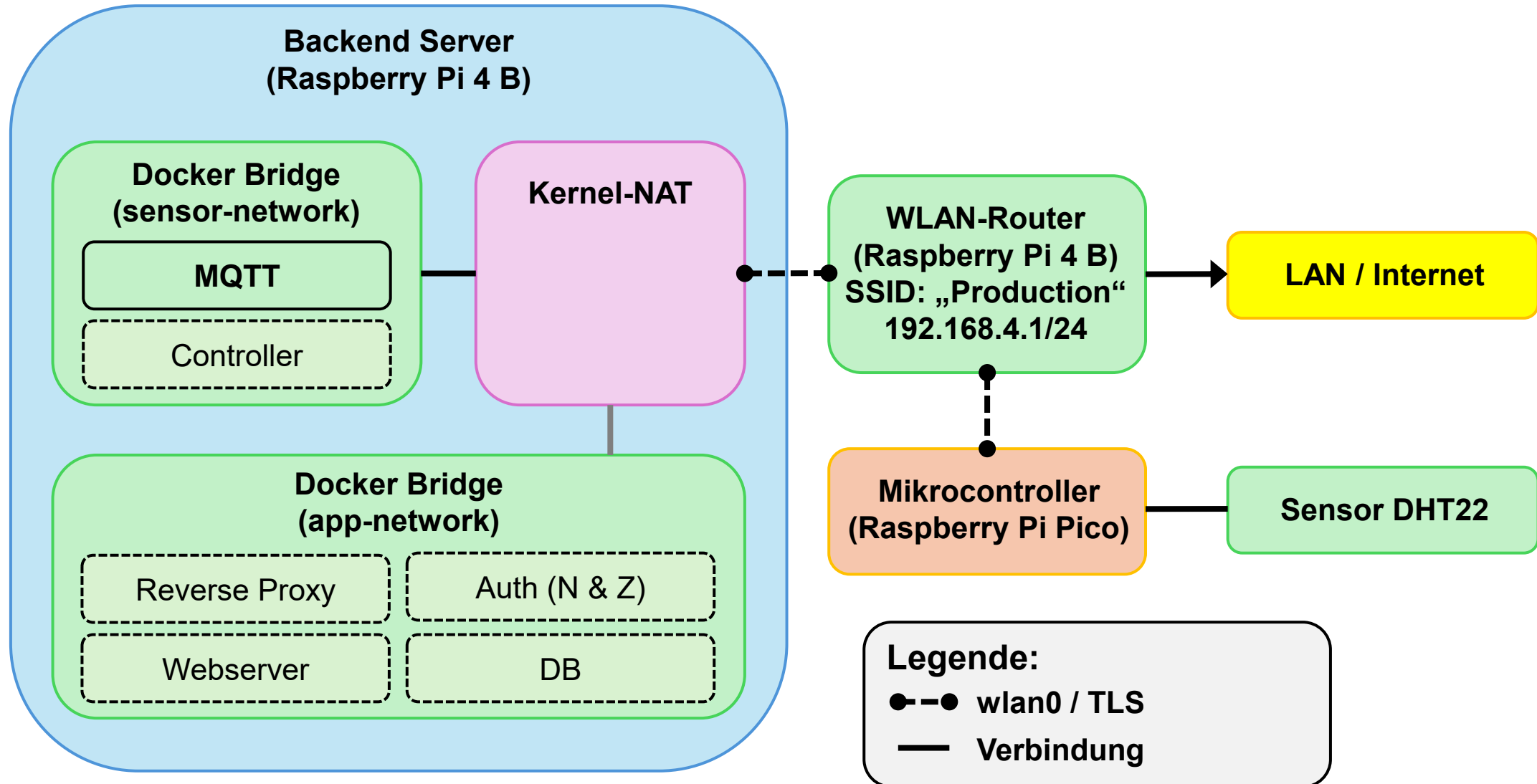
Test der Integration

- MQTT-Client verwenden
- mosquitto_sub
- mosquitto_pub
- Korrektheit prüfen

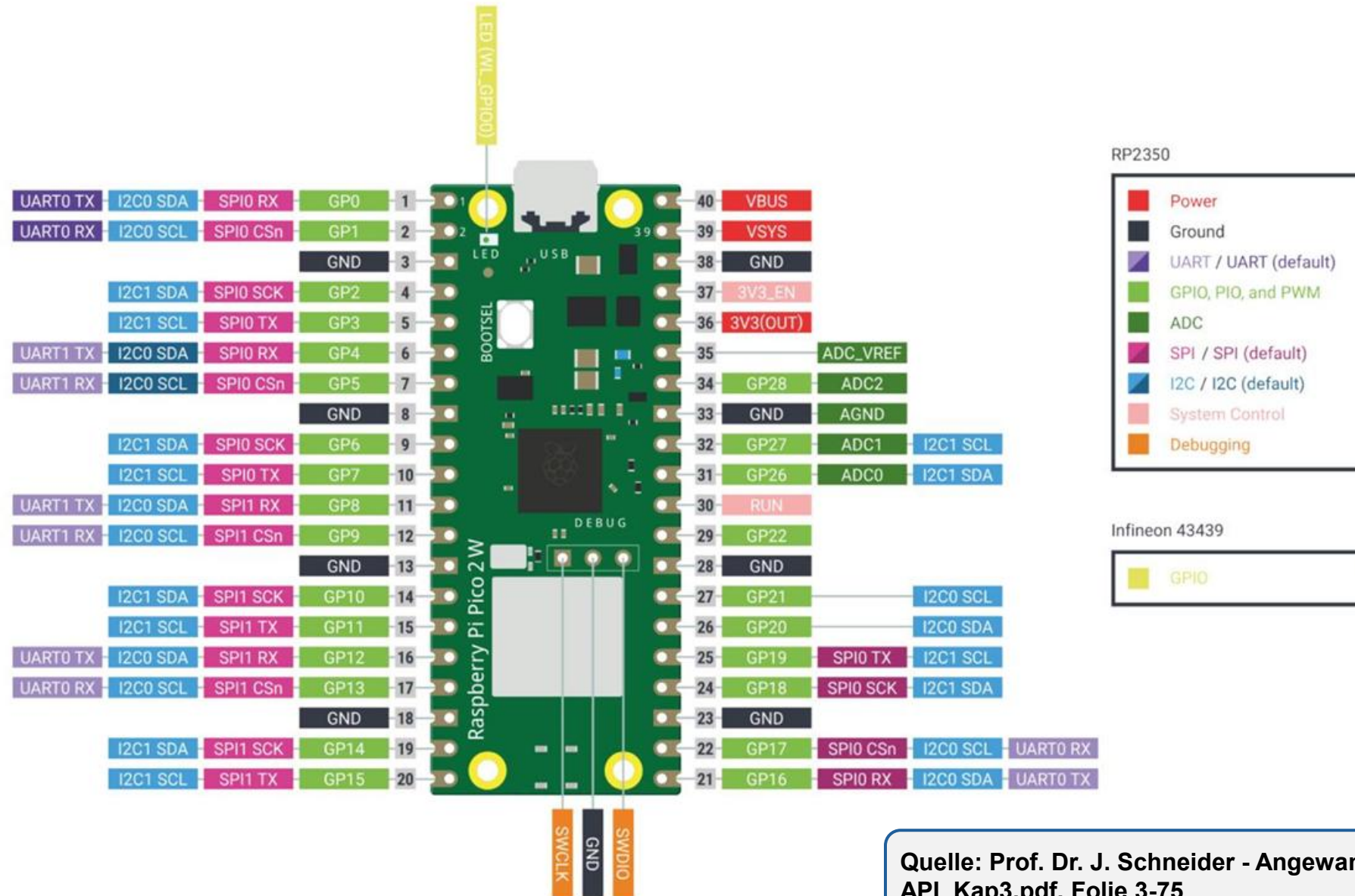
Aufgabenteilung



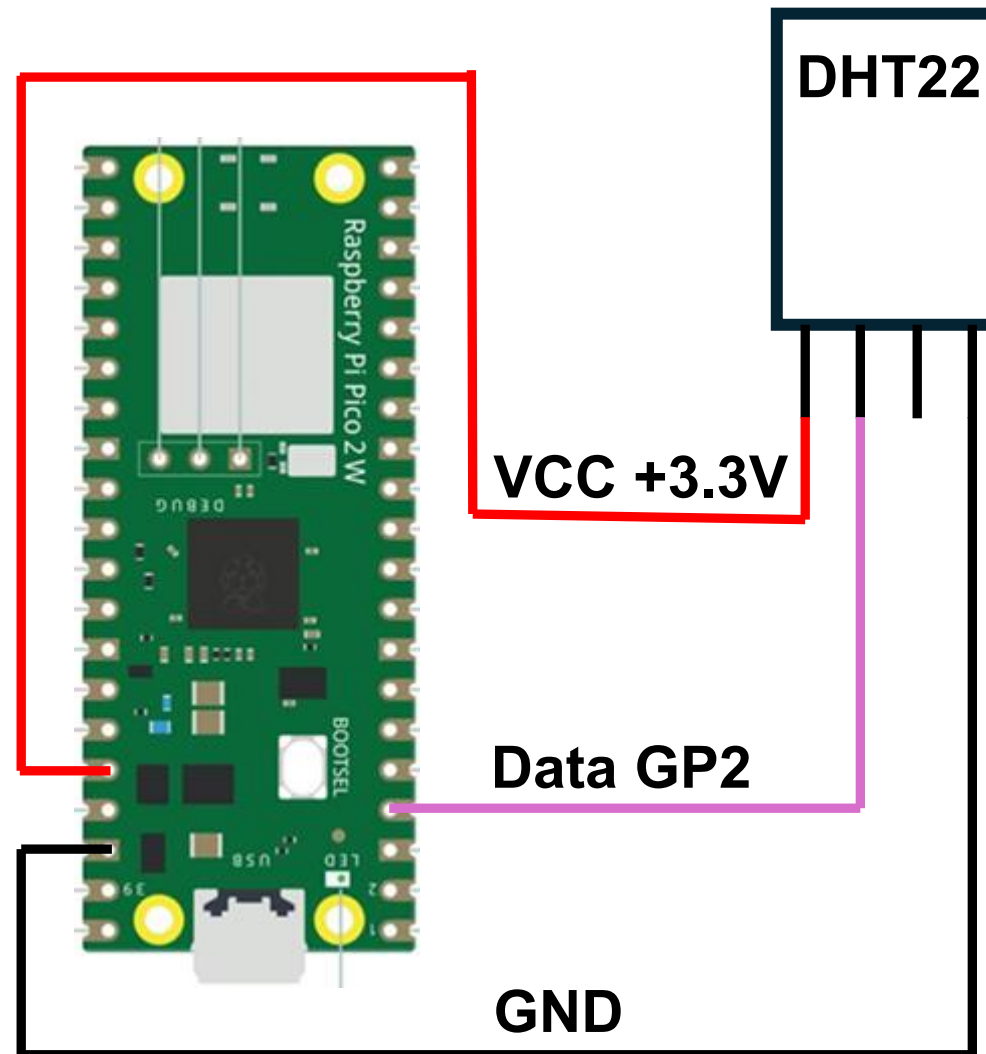
Deployment Diagramm



2. Hardware



2. Hardware – DHT-Sensor



2. Hardware

Pico Ausschnitt main.py

```
from time import sleep, time
import ujson
import wifi
import states
import mqtt

wlan = wifi.connect_wifi()

mqtt.client.connect()
```

2. Hardware – Pico Mikro Python Loop

```
while True:
    """Haupt-Loop"""
    try:
        states.sensor.measure()
        temp = states.sensor.temperature()
        hum = states.sensor.humidity()

        payload = ujson.dumps({
            "temperature": temp,
            "humidity": hum
        })

        mqtt.client.publish("sensor/data", payload)

        temp_state = states.determine_temp_state(temp)
        hum_state = states.determine_hum_state(hum)

        states.apply_state(temp_state, hum_state)

    except OSError as e:
        print("Failed to read sensor:", e)

    sleep(60)
```

2. Hardware

Pico Ausschnitt states.py

```
from picozero import pico_led
from machine import Pin
import dht

sensor = dht.DHT22(Pin(2, Pin.IN, Pin.PULL_UP))
radiator = Pin(12, Pin.OUT)    # Heizung
fan = Pin(13, Pin.OUT)        # Lüfter / Kühlung
```

3. Docker – docker-compose.yml

```
mqtt:
  image: eclipse-mosquitto:latest
  container_name: mqtt
  ports:
    - "8883:8883"
  volumes:
    - ./mosquitto/config:/mosquitto/config:rw
    - ./mosquitto/data:/mosquitto/data:rw
    - ./mosquitto/log:/mosquitto/log/:rw
    - /etc/timezone:/etc/timezone:ro
    - /set/localtime:/etc/localtime:ro
  restart: unless-stopped
  networks:
    sensor-network:
      aliases:
        - mqtt.local

networks:
  app-network:
    driver: bridge
  sensor-network:
    driver: bridge
```

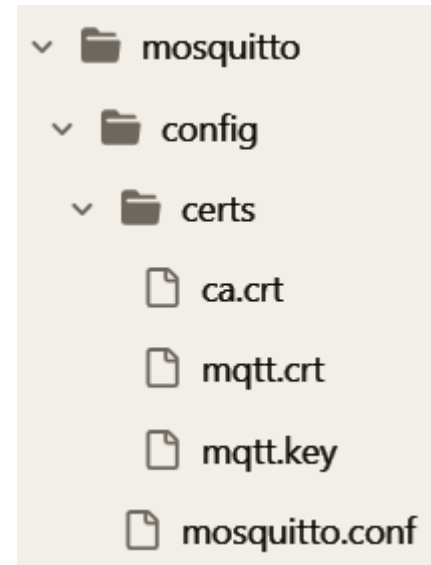
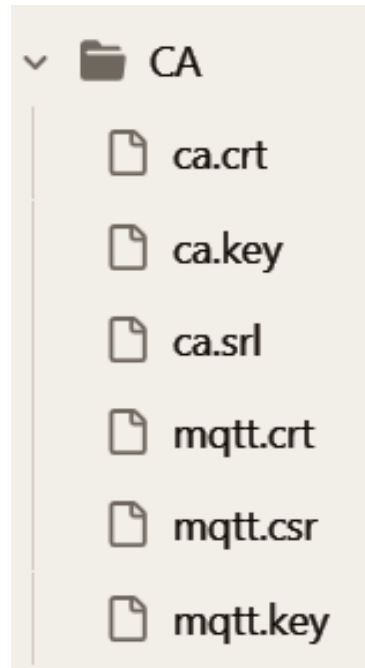
3. Docker

Docker compose

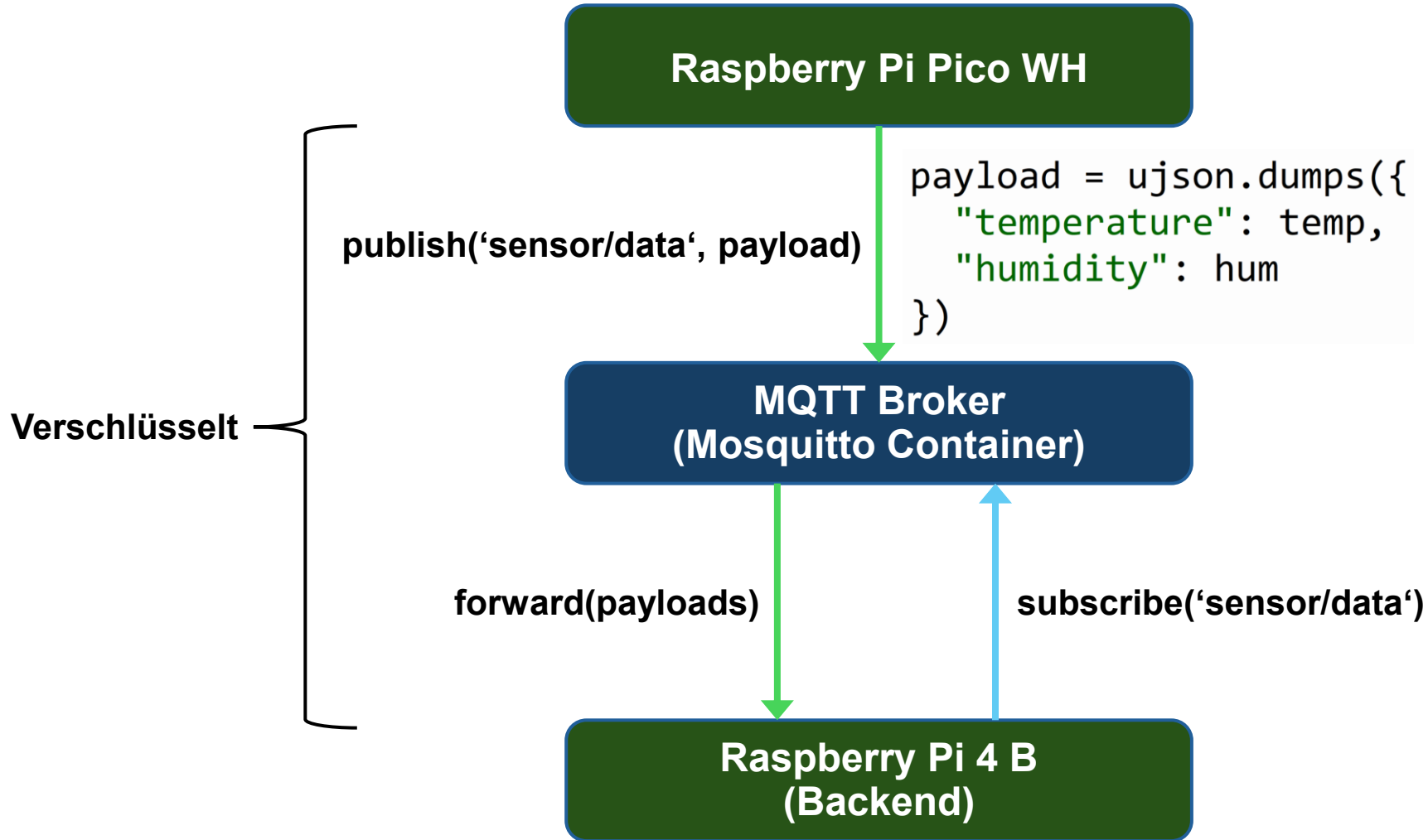
```
pi@Backend:~/Anwendungsprojekt.Inf $ sudo docker compose up --force-recreate
[+] up 5/5
✓Network anwendungsprojektinf_app-network      Created
✓Network anwendungsprojektinf_sensor-network  Created
✓Container mariadb                             Created
✓Container mqtt                                Created
✓Container backend                             Created
```

4. CA

CA und mosquitto Struktur



5. MQTT



5. MQTT – mosquitto.conf

```
# mosquitto.conf

# Port for TLS-connection
# Used protocol mqtt
listener 8883
protocol mqtt

#Client needs certificate
require_certificate true

# Disable anonymous access
allow_anonymous false
password_file /mosquitto/config/secure/pwfile

# activate authorization
acl_file /mosquitto/config/secure/acl

#Path to certificates for TLS connection in
# container
cafile /mosquitto/config/certs/ca.crt
certfile /mosquitto/config/certs/mqtt.crt
keyfile /mosquitto/config/certs/mqtt.key

#CN field for client not needed
use_identity_as_username false
```

5. MQTT

acl

```
user iotuser  
topic write sensor/data  
topic read actuator/control  
  
user testuser  
topic read sensor/data  
topic read actuator/control
```

5. MQTT – Publish / Subscribe

```
pi@Backend:~/Anwendungsprojekt.Inf $ mosquitto_pub -t 'sensor/data'
-m '23.9' -u iotuser -P [REDACTED] -p 8883 --cafile CA/ca.crt
```

```
pi@Backend:~/Anwendungsprojekt.Inf $ mosquitto_sub -t sensor/data
-u testuser -P [REDACTED] -p 8883 --cafile CA/ca.crt
23.9
23.9
23.9
23.9
```

```
pi@Backend:~/Anwendungsprojekt.Inf $ mosquitto_sub -t sensor/data
-u testuser -P [REDACTED] -p 8883 --cafile CA/ca.crt
{"humidity": 46.7, "temperature": 23.2}
{"humidity": 46.600004, "temperature": 23.2}
{"humidity": 46.7, "temperature": 23.2}
{"humidity": 46.7, "temperature": 23.2}
```

5. MQTT

Pico mqtt.py

```
import ssl
from umqtt.robust import MQTTClient

client = MQTTClient(
    client_id="pico",
    server="192.168.4.18",
    port=8883,
    user="iotuser",
    password=[REDACTED],
    ssl=True,
    ssl_params={
        "cert_reqs": ssl.CERT_REQUIRED,
        "ca_certs": "/ca.crt"
    }
)
```